

Name: _____

1. Definitions and Theorems	
2. Question 2	
3. Short Answer	
4. True or False	
5. Question 5	
6. Explanations	
TOTAL	
5. Isomorphism	
6. Normality	

1. (20 points) Definitions and Theorems: Complete the following definitions or theorems.

(a) Theorem: Let G and H be finite cyclic groups. Then $G \oplus H$ is cyclic if and only if:

(b) Definition: A group G is the internal direct product of H and K ($G = H \times K$) if:

(c) Definition: An *isomorphism* ϕ from a group G to a group $\bar{G}$ is:

(d) Definition: A subgroup H of a group G is called a normal subgroup of G if:

(e) State Cauchy's Theorem for Finite Abelian Groups:

2. (15 points) Consider $G = \mathbb{Z}_{18}/\langle 6 \rangle$. Justify your answers for each of the following.

(a) What is the order of G ? _____

(b) What is the order of $14 + \langle 6 \rangle$ in G ? _____

(c) What is the inverse of $14 + \langle 6 \rangle$ in G ? _____

(d) Is G cyclic? Explain.

3. (25 points, 5 points each) Short Answer: For full credit, you must justify your answer by providing clear work or a very brief explanation.

(a) How many cyclic subgroups of order 10 are in $\mathbb{Z}_{100} \oplus \mathbb{Z}_{25}$? _____

(b) How many automorphisms are there of $\mathbb{Z}_{18}$? _____

(c) If H and K are subgroups of G , where $|H| = 36$ and $|K| = 28$, what are the possible orders of $H \cap K$? _____

(d) In $\mathbb{Z}$, Let $H = \langle 15 \rangle$ and $K = \langle 18 \rangle$. Does $G = HK$? _____

(e) Let $G = U(32)$ and let $H = \{1, 31\}$. Which of the following is G/H isomorphic to: $\mathbb{Z}_2 \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_2$, $\mathbb{Z}_2 \oplus \mathbb{Z}_4$, or $\mathbb{Z}_8$? _____

4. (27 points) True or False. Circle True if the statement is always true. Otherwise, circle False.

True False If G is a group of order 40 and H is a subgroup of G with order 20, then H is normal in G .

True False If G/H is Abelian, then G is Abelian.

True False If G is a cyclic group, then every subgroup of G is normal.

True False S_6 does not have a subgroup of order 7.

True False The cosets of $\langle 4 \rangle$ in $\mathbb{Z}_{12}$ are $\{0, 1, 2, 3\}$, $\{4, 5, 6, 7\}$, and $\{8, 9, 10, 11\}$.

True False If the order of a group G is 8, then G is isomorphic to a subgroup of S_8 .

True False Let G be a group such that d divides $|G|$. Then G has a subgroup of order d .

True False Let $H = \{0, \pm 4, \pm 8, \pm 12, \dots\}$. Then the cosets $-2 + H$ and $6 + H$ are equal.

True False The groups $U(5)$ and $\mathbb{Z}_4$ are isomorphic.

5. (12 points) Answer the following about $U(72)$.

(a) Write $U(72)$ as the external direct product of cyclic groups. _____

(b) What is the order of $U(72)$? _____

(c) Is $U(72)$ cyclic? Why or why not?

(d) How many elements of order 12 are in $U(72)$? _____

- Page 5

7. **Isomorphism Proof:** Prove that $\mathbb{Z}$ is isomorphic to $n\mathbb{Z} = \{\dots - 3n, -2n, -n, 0, n, 2n, 3n, \dots\}$ for $n \neq 0$.

8. **Normality Proof:** If N and M are normal subgroups of G , prove that NM is also a normal subgroup of G .